

COMMANDS

1

```

jell@DESKTOP-GM2LUJJI:~/OSLab$ uname -a
Linux DESKTOP-GM2LUJJI 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux

```

2

```

all@kali:~/Kali$ sudo dmidecode -s 4
# dmidecode 3.11
#0000:00000000-00000000-00000000-00000000-00000000-00000000-00000000-00000000
Processor Information:
  Processor: Intel(R) Core(TM) i5-3210M CPU @ 2.50GHz
  Processor ID: 0-3
  Vendor ID: GenuineIntel
  Model name: Intel(R) Core(TM) i5-3210M CPU @ 2.50GHz
  CPU family: 6
  Model: 58
  Thread(s) per core: 2
  Core(s) per socket: 2
  Socket(s): 1
  Stepping: 9
  Bogomips: 4983.00
  Flags: fpu vme de pse tsc msr pae mce cx8
apic sep mtrr pge mca cmov pat ps
all nx rdtscln constant_t_ sse4_2 popcnt aes xsave aux f16c r
lqy cpuid tsc_known_freq pni pclm e36 cflush mmx fxsr sse sse2 ss h
ulqq sse3e cx16 pdcm pcid sse4_1
t sys
  ibrs ibpb stibp fsgsbase smep erms
drand hypervisorlahf_lm pti ssbd
xsavopt mve_clear flush_lid arch_
capabilities
Virtualization features:
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 64 KiB (2 instances)
  L1i: 64 KiB (2 instances)
  L2: 512 KiB (2 instances)
  L3: 3 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-3
Vulnerabilities:
  Gather data sampling: Not affected
  GDSbwrite: Not affected
  Indirect target selection: Mitigation; Aligned branch/return

```

3

```
Well@DESKTOP-GM2LUJI:~/OSLab$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0    1G    0 disk [SWAP]
sdd    8:48   0     1T    0 disk /mnt/wslg/distro
/
```

4

```

root@kali:~# cat /etc/passwd | grep -v ^$ | sort -t: -k1,1 -k2,2n -k3,3n -k4,4n -k5,5n -k6,6n -k7,7n -k8,8n -k9,9n -k10,10n -k11,11n -k12,12n -k13,13n -k14,14n -k15,15n -k16,16n -k17,17n -k18,18n -k19,19n -k20,20n -k21,21n -k22,22n -k23,23n -k24,24n -k25,25n -k26,26n -k27,27n -k28,28n -k29,29n -k30,30n -k31,31n -k32,32n -k33,33n -k34,34n -k35,35n -k36,36n -k37,37n -k38,38n -k39,39n -k40,40n -k41,41n -k42,42n -k43,43n -k44,44n -k45,45n -k46,46n -k47,47n -k48,48n -k49,49n -k50,50n -k51,51n -k52,52n -k53,53n -k54,54n -k55,55n -k56,56n -k57,57n -k58,58n -k59,59n -k60,60n -k61,61n -k62,62n -k63,63n -k64,64n -k65,65n -k66,66n -k67,67n -k68,68n -k69,69n -k70,70n -k71,71n -k72,72n -k73,73n -k74,74n -k75,75n -k76,76n -k77,77n -k78,78n -k79,79n -k80,80n -k81,81n -k82,82n -k83,83n -k84,84n -k85,85n -k86,86n -k87,87n -k88,88n -k89,89n -k90,90n -k91,91n -k92,92n -k93,93n -k94,94n -k95,95n -k96,96n -k97,97n -k98,98n -k99,99n -k100,100n -k101,101n -k102,102n -k103,103n -k104,104n -k105,105n -k106,106n -k107,107n -k108,108n -k109,109n -k110,110n -k111,111n -k112,112n -k113,113n -k114,114n -k115,115n -k116,116n -k117,117n -k118,118n -k119,119n -k120,120n -k121,121n -k122,122n -k123,123n -k124,124n -k125,125n -k126,126n -k127,127n -k128,128n -k129,129n -k130,130n -k131,131n -k132,132n -k133,133n -k134,134n -k135,135n -k136,136n -k137,137n -k138,138n -k139,139n -k140,140n -k141,141n -k142,142n -k143,143n -k144,144n -k145,145n -k146,146n -k147,147n -k148,148n -k149,149n -k150,150n -k151,151n -k152,152n -k153,153n -k154,154n -k155,155n -k156,156n -k157,157n -k158,158n -k159,159n -k160,160n -k161,161n -k162,162n -k163,163n -k164,164n -k165,165n -k166,166n -k167,167n -k168,168n -k169,169n -k170,170n -k171,171n -k172,172n -k173,173n -k174,174n -k175,175n -k176,176n -k177,177n -k178,178n -k179,179n -k180,180n -k181,181n -k182,182n -k183,183n -k184,184n -k185,185n -k186,186n -k187,187n -k188,188n -k189,189n -k190,190n -k191,191n -k192,192n -k193,193n -k194,194n -k195,195n -k196,196n -k197,197n -k198,198n -k199,199n -k200,200n -k201,201n -k202,202n -k203,203n -k204,204n -k205,205n -k206,206n -k207,207n -k208,208n -k209,209n -k210,210n -k211,211n -k212,212n -k213,213n -k214,214n -k215,215n -k216,216n -k217,217n -k218,218n -k219,219n -k220,220n -k221,221n -k222,222n -k223,223n -k224,224n -k225,225n -k226,226n -k227,227n -k228,228n -k229,229n -k230,230n -k231,231n -k232,232n -k233,233n -k234,234n -k235,235n -k236,236n -k237,237n -k238,238n -k239,239n -k240,240n -k241,241n -k242,242n -k243,243n -k244,244n -k245,245n -k246,246n -k247,247n -k248,248n -k249,249n -k250,250n -k251,251n -k252,252n -k253,253n -k254,254n -k255,255n -k256,256n -k257,257n -k258,258n -k259,259n -k260,260n -k261,261n -k262,262n -k263,263n -k264,264n -k265,265n -k266,266n -k267,267n -k268,268n -k269,269n -k270,270n -k271,271n -k272,272n -k273,273n -k274,274n -k275,275n -k276,276n -k277,277n -k278,278n -k279,279n -k280,280n -k281,281n -k282,282n -k283,283n -k284,284n -k285,285n -k286,286n -k287,287n -k288,288n -k289,289n -k290,290n -k291,291n -k292,292n -k293,293n -k294,294n -k295,295n -k296,296n -k297,297n -k298,298n -k299,299n -k300,300n -k301,301n -k302,302n -k303,303n -k304,304n -k305,305n -k306,306n -k307,307n -k308,308n -k309,309n -k310,310n -k311,311n -k312,312n -k313,313n -k314,314n -k315,315n -k316,316n -k317,317n -k318,318n -k319,319n -k320,320n -k321,321n -k322,322n -k323,323n -k324,324n -k325,325n -k326,326n -k327,327n -k328,328n -k329,329n -k330,330n -k331,331n -k332,332n -k333,333n -k334,334n -k335,335n -k336,336n -k337,337n -k338,338n -k339,339n -k340,340n -k341,341n -k342,342n -k343,343n -k344,344n -k345,345n -k346,346n -k347,347n -k348,348n -k349,349n -k350,350n -k351,351n -k352,352n -k353,353n -k354,354n -k355,355n -k356,356n -k357,357n -k358,358n -k359,359n -k360,360n -k361,361n -k362,362n -k363,363n -k364,364n -k365,365n -k366,366n -k367,367n -k368,368n -k369,369n -k370,370n -k371,371n -k372,372n -k373,373n -k374,374n -k375,375n -k376,376n -k377,377n -k378,378n -k379,379n -k380,380n -k381,381n -k382,382n -k383,383n -k384,384n -k385,385n -k386,386n -k387,387n -k388,388n -k389,389n -k390,390n -k391,391n -k392,392n -k393,393n -k394,394n -k395,395n -k396,396n -k397,397n -k398,398n -k399,399n -k400,400n -k401,401n -k402,402n -k403,403n -k404,404n -k405,405n -k406,406n -k407,407n -k408,408n -k409,409n -k410,410n -k411,411n -k412,412n -k413,413n -k414,414n -k415,415n -k416,416n -k417,417n -k418,418n -k419,419n -k420,420n -k421,421n -k422,422n -k423,423n -k424,424n -k425,425n -k426,426n -k427,427n -k428,428n -k429,42
```

5

```
dell@DESKTOP-GM2LUJI:~/OSLab$ top
top - 15:11:48 up 2:33, 1 user, load average: 0.05, 0.01, 0
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0
%Cpu(s): 2.2 us, 2.2 sy, 0.0 ni, 95.7 id, 0.0 wa, 0.0 hi,
MiB Mem : 3868.6 total, 3291.4 free, 502.4 used, 223
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3366
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	root	20	0	24484	14312	10164	S	0.0	0.4
2	root	20	0	3180	2212	2072	S	0.0	0.1
8	root	20	0	3180	2048	1940	S	0.0	0.1
124	root	20	0	2888	2012	1884	S	0.0	0.1
125	root	20	0	4472	3120	2876	S	0.0	0.1
126	message+	20	0	9056	5768	4752	S	0.0	0.1
157	root	20	0	44096	29816	14672	S	0.0	0.8
167	root	20	0	18440	9492	8192	S	0.0	0.2
178	syslog	20	0	220548	6008	4712	S	0.0	0.2
226	_chrony	20	0	23872	11368	7064	S	0.0	0.3
232	_chrony	20	0	12096	2508	1868	S	0.0	0.1
270	root	20	0	123460	32464	17852	S	0.0	0.8
279	root	20	0	5492	2828	2644	S	0.0	0.1
329	root	20	0	3184	1116	980	S	0.0	0.0
331	root	20	0	3200	1252	1108	S	0.0	0.0
333	dell	20	0	6268	5672	3992	S	0.0	0.1
337	root	20	0	8644	5524	4684	S	0.0	0.1
397	dell	20	0	22328	9688	7300	S	0.0	0.2
401	dell	20	0	22920	4276	2272	S	0.0	0.1
431	dell	20	0	6136	5548	3992	S	0.0	0.1
721	polkitd	20	0	306444	8512	7624	S	0.0	0.2
1448	root	19	-1	50604	15572	14304	S	0.0	0.4
1511	root	20	0	33916	9284	7484	S	0.0	0.2
1583	systemd+	20	0	22416	12364	9868	S	0.0	0.3
2343	dell	20	0	8676	4952	4384	S	0.0	0.1
2347	dell	20	0	380948	7932	7276	S	0.0	0.2
2354	dell	20	0	8540	4656	4284	S	0.0	0.1
2355	dell	20	0	476868	17820	14924	S	0.0	0.4
2360	dell	20	0	304804	6544	6044	S	0.0	0.2

6

```
dell@DESKTOP-GM2LUJI:~/OSLab$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 36 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 4
On-line CPU(s) list: 0-3
Vendor ID: GenuineIntel
Model name: Intel(R) Core(TM) i5-3210M CPU @ 2.50GHz
CPU family: 6
Model: 58
Thread(s) per core: 2
Core(s) per socket: 2
Socket(s): 1
Stepping: 9
BogoMIPS: 4983.80
Flags: fpu vme de pse tsc msr pae mce cx8 c arch_perfmon rep_good nopl xtopo apic sep mtrr pge mca cmov pat ps e36 clflush mmx fxsr sse sse2 ss h t sysc
all nx rdtscp lm constant_tss sse4_1 popcnt aes xsave avx f16c r drand hypervisorlahf_lm pti ssbd ui0dq sse3 cx16 pdcm pcid sse4_1
Virtualization features:
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 64 KiB (2 instances)
L1i: 64 KiB (2 instances)
L2: 512 KiB (2 instances)
L3: 3 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-3
Vulnerabilities:
gather data sampling: Not affected
ghostwrite: Not affected
Indirect target selection: Mitigation; Aligned branch/return
Itlb multihit: Mitigation; VMX unsupported
L1tf: Mitigation; PTE Inversion
Mds: Mitigation; Clear CPU buffers; SMT
Host state unknown
```

Activate Windows
Go to Settings to activate Windows.

7

```
dell@DESKTOP-GM2LUJI:~/OSLab$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	3.8Gi	502Mi	3.2Gi	4.4Mi	223Mi	3.3Gi
Swap:	1.0Gi	0B	1.0Gi			

CODING OUTPUTS

1

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./Oneprocess  
PID = 4977
```

2

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./OneProcess1  
Hello  
Hello  
dell@DESKTOP-GM2LUJI:~/OSLab$
```

3

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./PC1  
I am Child  
I am Parent
```

4

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./PIDs  
Parent PID = 5144  
Child PID = 5145  
Child PID = 5145  
Parent PID = 331
```

5

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./P5  
Child Executing  
Parent Executing
```

6

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./p6  
PID = 5176  
PID = 5178  
PID = 5177  
PID = 5179
```

7

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./p7
Parent Waiting...
Child executing ls...
dell@DESKTOP-GM2LUJI:~/OSLab$ total 208
-rw-r--r-- 1 dell dell    0 Jul 27 06:03 Compiler
drwxr-xr-x 7 dell dell  4096 Aug 12 07:07 OSSP_Project
-rw-r--r-- 1 dell dell    99 Aug 12 15:23 OneProcess.c
-rwxr-xr-x 1 dell dell 16000 Jul 29 06:44 OneProcess1
-rw-r--r-- 1 dell dell    95 Jul 29 06:44 OneProcess1.c
-rwxr-xr-x 1 dell dell 16000 Aug 12 15:24 Oneprocess
-rwxr-xr-x 1 dell dell 16032 Jul 29 06:51 P5
-rw-r--r-- 1 dell dell   215 Jul 29 06:51 P5.c
-rwxr-xr-x 1 dell dell 15992 Jul 29 06:47 PC1
-rw-r--r-- 1 dell dell   164 Jul 29 06:46 PC1.c
-rwxr-xr-x 1 dell dell 16080 Jul 29 06:48 PIDs
-rw-r--r-- 1 dell dell   278 Jul 29 06:48 PIDs.c
-rwxr-xr-x 1 dell dell 15952 Jul 27 06:02 hello
-rw-r--r-- 1 dell dell    79 Jul 27 06:01 hello.c
-rwxr-xr-x 1 dell dell 16040 Jul 29 06:56 p10
-rw-r--r-- 1 dell dell   132 Jul 29 06:55 p10.c
-rwxr-xr-x 1 dell dell 16392 Jul 29 06:57 p11
-rw-r--r-- 1 dell dell   628 Jul 29 06:56 p11.c
-rwxr-xr-x 1 dell dell 16032 Jul 29 06:52 p6
-rw-r--r-- 1 dell dell   117 Jul 29 06:52 p6.c
-rwxr-xr-x 1 dell dell 16032 Jul 29 06:54 p7
-rw-r--r-- 1 dell dell   220 Jul 29 06:54 p7.c
ls
```

10

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./p10
PID = 5221
PID = 5222
PID = 5220
PID = 5224
PID = 5225
PID = 5223
PID = 5227
PID = 5226
```

11

```
dell@DESKTOP-GM2LUJI:~/OSLab$ ./p11
Enter Linux Command: ls

Parent Process
Parent PID : 5237
Child PID : 5238

Child Process
Child PID : 5238
Parent PID: 5237
Compiler  OSSP_Project  OneProcess.c  OneProcess1  OneProcess1.c  Oneprocess  P5  P5.c  PC1  PC1.c  PIDs  PIDs.c  hello
hello.c  p10  p10.c  p11  p11.c  p6  p6.c  p7  p7.c

Child Process Completed
dell@DESKTOP-GM2LUJI:~/OSLab$
```